Question	Answer	Marks	Partial Marks
2(c)	4	**2**	**M1** for $\frac{100+2.5}{100}\times[...]=\frac{100+6.6}{100}$ oe
2(d)(i)	72.3 or 72.31...	**2**	**M1** for $80\times\left(\frac{100-2}{100}\right)^5$ oe
2(d)(ii)	4 nfww	**3**	**B2** for answer 9 nfww or **M2** for correct trials with values giving either side of 67 or **M1** for $80\times\left(\frac{100-2}{100}\right)^n=67$ or $their(i)\times\left(\frac{100-2}{100}\right)^k=67$ or an evaluated trial with $n \geqslant 6$ or $k \geqslant 1$

Question	Answer	Marks	Partial Marks
3(a)(i)	6	**3**	**B2** for $4x + 6 = 30$ or better or **M1** for $x + x + 7 + 2x - 1$ [= 30]
3(a)(ii)	21	**3**	**M2** for (555 – *their* x × 15 – *their* (x + 7) × 18) ÷ *their* (2x – 1) or **M1** for *their* x × 15 or *their* (x + 7) × 18
3(b)(i)	8	**2**	**M1** for isolating the term in w or correctly removing all fractions e.g. $\frac{3w}{16}=1+\frac{1}{2}$ or better or 3w – 16 = 8
3(b)(ii)	–3	**2**	**M1** for $2^{-y}=8$ or $2^{y}=\frac{1}{8}$ or $2^{-y}=their\ w$ or better

Question	Answer	Marks	Partial Marks
5(a)(i)(a)	$\frac{(8-2)\times 180}{8\times 2}$ oe	**M2**	**M1** for $\frac{(8-2)\times 180}{8}$ or $\frac{360}{8}$ or $\frac{(2\times 8-4)\times 90}{8}$
5(a)(i)(b)	174 or 173.8....	**4**	**M3** for $\frac{1}{2}\times 6\times OM$ oe or $\frac{1}{2}\times (OA)^2\times \sin 45$ oe or $\frac{1}{2}\times 6\times OA\times sin67.5$ oe where OA and OM are as in the **M2** or **M2** for $OM = 3\times \tan 67.5$ oe or for $OA = \left(\frac{3}{cos67.5}\right)$ or $\frac{6\times sin67.5}{sin45}$ oe or **M1** for $\frac{OM}{3} = \tan 67.5$ oe or for $\frac{3}{OA} = cos67.5$ oe or for $\frac{sin45}{6} = \frac{sin67.5}{OA}$ oe
5(a)(ii)	193 or 193.0 to 193.1	**3**	**M2** for $\pi\times\left(\frac{3}{cos67.5}\right)^2$ oe or **M1** for $\frac{3}{r} = cos67.5$ or $\frac{sin45}{6} = \frac{sin67.5}{r}$

Question	Answer	Marks	Partial Marks
7(a)	39.6 or 39.57....	**4**	**M2** for [cos =] $\frac{14^2+12^2-9^2}{2\times14\times12}$ or **M1** for $9^2 = 14^2 + 12^2 - 2 \times 14 \times 12 \times \cos ACD$ **A1** for 0.7708... or 0.771 or $\frac{37}{48}$ oe
7(b)	$\frac{14\sin 25}{\sin 123}$	**M2**	**M1** for $\frac{\sin 123}{14} = \frac{\sin 25}{BC}$ oe
	7.054…	**A1**	
7(c)	3.74 or 3.735 to 3.739	**3**	**M2** for 7.05 × sin 32 or **M1** for recognition that the line from *B* is perpendicular to *AC*
7(d)	11.8 or 11.83 to 11.85	**4**	**M1** for 32 + *their*(a) soi **M2** for $12^2 + 7.05^2 - 2\times12\times7.05\times\cos(their(a)+32)$ or **M1** for $\cos(their(a)+32) = \frac{12^2+7.05^2-BD^2}{2\times12\times7.05}$
7(e)	309.6 or 309.57...	**2**	**FT** 270 + *their(a)* **M1** for 270 + *their(a)* oe

Question	Answer	Marks	Partial Marks
9(a)	$x^2 - x - 30 [= 0]$	**B3**	**M1** for $(2x+1)(x-1) - x^2 = 29$ oe **B1** for $(2x+1)(x-1) = 2x^2 - 2x + x - 1$ oe soi
	$(x-6)(x+5)$ oe	**M1**	or correct factors for *their* 3 term quadratic equation or for correct substitution into quadratic formula or correctly completing the square for *their* 3 term quadratic equation
	$x = 6$ cao	**B1**	
	12 or 2 × *their x* evaluated or $k = 2x$ stated	**B1 FT**	
9(b)(i)	$(y+1)^3 - y^3 = 5$ oe	**M1**	
	$(y+1)^3 = y^3 + 3y^2 + 3y + 1$ soi	**B2**	**B1** for $(y+1)^2 = y^2 + y + y + 1$ oe soi
	Completion to $3y^2 + 3y - 4 = 0$	**A1**	With no errors or omissions
9(b)(ii)	$\dfrac{-3 \pm \sqrt{3^2 - 4(3)(-4)}}{2 \times 3}$	**B2**	or **B1** for $\sqrt{3^2 - 4(3)(-4)}$ or for $\dfrac{-3 + \sqrt{...}}{2 \times 3}$ or $\dfrac{-3 - \sqrt{...}}{2 \times 3}$
	0.44	**B2**	**B1** for 0.758 or 0.7583...